Diagnosis of Ear Conditions Using Deep Learning Approach

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***Abstract*— The lack of medical expertise and facilities makes it difficult to diagnose the diseases and the misdiagnosis can severely affect the health of the patients. This problem can be overcome by automating the process of diagnosis using artificial intelligence algorithms like deep learning. It has been used a lot for developing methods for the diagnosis of diseases with help of medical images. The CNN performs quite well for the diagnosis of diseases using deep learning and has been used for the diagnosis of a lot of diseases. Ear diseases are one of the most common problems that are encountered. They can be easily treated with proper diagnosis and medication using deep learning. This work proposes a deep learning-based method employing CNN for the detection of ear infections. It is capable of classifying ear images belonging to four categories (Chronic otitis media, Earwax plug, Myringosclerosis and Normal) and obtained training and validation accuracies of 99.98% and 97.37% respectively.**

Keywords*— *Myringosclerosis, Chronic Otitis Media, Earwax-Plug, Convolutional Neural Network, Deep Learning

I. INTRODUCTION

Ear diseases are one of the most commonly occurring diseases. The common symptoms of ear diseases are irritation, inflammation, earache, fluid draining from the ear, tugging of the ear, and hearing strange sounds. Many ear infections can be treated easily with timely diagnosis and proper medication. Chronic otitis media represents problems occurring in the middle ear like recurring middle ear infections and hole in the eardrum. Ear wax is a brownish wax like substance that is secreted in the ear canal of mammals. It protects the ear canal from microorganisms, insects and water and helps in its cleaning and lubrication. It is also known as cerumen. Earwax plug (cerumen impaction) is clogging of the outer ear canal caused by tightly packed cerumen. Myringosclerosis is characterized by calcification of the tympanic membrane and may restrict its mobility.

Deep learning has been widely used for detecting various diseases using medical images. Some of the instances of its application for detecting diseases have been listed below.

Mohammad Azam Khan et al. [1] used convolutional neural networks (CNN) for detecting tympanic membrane and middle ear infections and an accuracy of 95% was achieved. Dongchul Cha et al. [2] used an ensemble classifier using Inception v-3 ResNet 101 for detecting several ear diseases. It achieved an average accuracy of 93.67% for the 5fold cross-validation. Pressler, M., et al. [3] used CNN to detect ear abnormalities using 2-dimensional images and achieved 100% training accuracy and 94.1% test accuracy. Adria Romero Lopez et al. [4] used a modified VGG-16 to classify skin lesions as benign and malignant. It achieved 78.66% sensitivity on the ISIC Archive dataset. Heba Mohsen et al. [5] developed method employing deep neural network using principal component analysis along with discrete wavelet transformation. Justin Ker et al. [6] reviewed various applications employing deep learning techniques in medical image analysis.

Turin Ozturk et al. [7] chose the Darknet-19 model for developing a model to detect COVID-19 using X-ray images. It obtained 98.08% accuracy while performing binary classification and 87.02% accuracy while performing multi-class classification. Siddharth Bhatia et al. [8] used UNet and ResNet for feature extraction and then fed them to various classifiers to detect lung cancer. The ensemble of UNet and Random forest obtained an accuracy of 84%. Bajwa, M.N et al. [9] used deep neural networks for the diagnosis of several skin diseases. 80% accuracy was achieved on the DermNet dataset and 93% average accuracy was reported on the ISIC archive.

H. Zhou et al. [10] employed CNN for the classification of skin diseases using dermoscopy images and obtained an accuracy of 90%. H. S. Baweja and T. Parhar [11] used an approach employing CNN for detecting leprosy lesions and achieved an accuracy of 91.6%. V. Srividhya et al. [12] a CNN based approach involving highly perceptive feature extraction for the diagnosis of melanoma. It achieved 95% identification efficiency and 93.3% sensitivity. Islam J. et al. [13] proposed a Deep CNN model for detecting Alzheimer's disease with the help of MRI.

Joachim Krois et al. [14] proposed a method employing deep learning for the detecting of periodontal bone loss with the help of panoramic dental radiographs. It achieved a classification accuracy of 81%. Amjad Rehman et al. [15] used AlexNet for the diagnosis of acute lymphoblastic leukemia and obtained 97.78% accuracy. Zhenwei Zhang et al. [16] proposed a model based on the single-shot multi-box detector for the automatic hyoid bone detection using fluoroscopic images. Muktabh Mayank Srivastava et al. [17] developed a computer-aided mechanism for detecting tooth

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caries using FCNN. Parnian Afshar et al. [18] used Capsule network for the detecting brain tumours to overcome the problem of the lack of availability of large datasets. Muhammed Talo et al. [19] used ResNet34 to detect brain abnormalities using MR images. It achieved an accuracy of 100%.

The main contribution of this proposed work is that it proposes an efficient and reliable deep learning method for detecting ear conditions. The dataset having images belonging to four classes (Normal, Chronic otitis media, Earwax Plug, and Myringosclerosis) to train the model. Various augmentation techniques have been applied to make the dataset larger to increase the reliability of the method and a CNN has been used to perform classification for the diagnosis of the ear conditions.

The remaining paper is organised as given below. Section II elaborates on the dataset employed and the proposed work. Section III presents the results. Section IV comprises the conclusion and the future work.

II. MATERIALS AND METHODOLOGY

The dataset was augmented and pre-processed. After pre-processing it was split into three parts for training, testing and validation. The training and validation datasets were served as an input to the CNN for training and then the trained model was tested using the testing dataset. This section elaborates the dataset, data pre-processing and the proposed CNN architecture. Fig. 1 shows the proposed method.

A. Dataset and Pre-processing

The dataset used in this paper is the Ear imagery database developed by the Universidad de Chile [20]. It consists of 880 images belonging to four classes (Normal, Chronic Otitis Media, Earwax Plug, and Myringosclerosis). Fig. 2 shows images belonging to the four classes.

The various data augmentation operations (rotation, zoom, horizontal flip, and vertical flip) were performed on the images. The ImageDataGenerator was employed for data augmentation. The zoom range was set to 0.1 and the rotation range was taken to be 20. 5703 images were obtained as the result of the augmentation. 3650 images were used to train the model. 1141 images were used for validation and remaining 1369 images were used for testing the model.

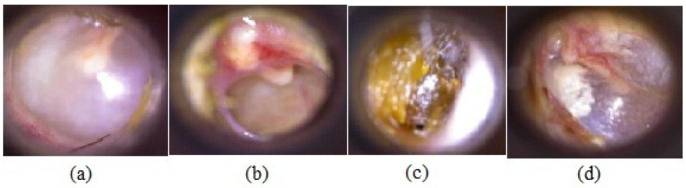

Fig 2. Images belonging to the four categories: (a) Normal (b) Chronic otitis media (c) Earwax plug (d)Myringosclerosis

The images were normalized by multiplying each pixel with a factor of 1/255. The images were resized to 247×286 before being fed to the network. One hot encoding was used to convert categorical labels into binary.

B. Proposed Convolutional Neural Network Architecture

A CNN is capable of automatically extracting features. It does not require manual feature extraction. The convolutional layers in CNN extract features from the input images.

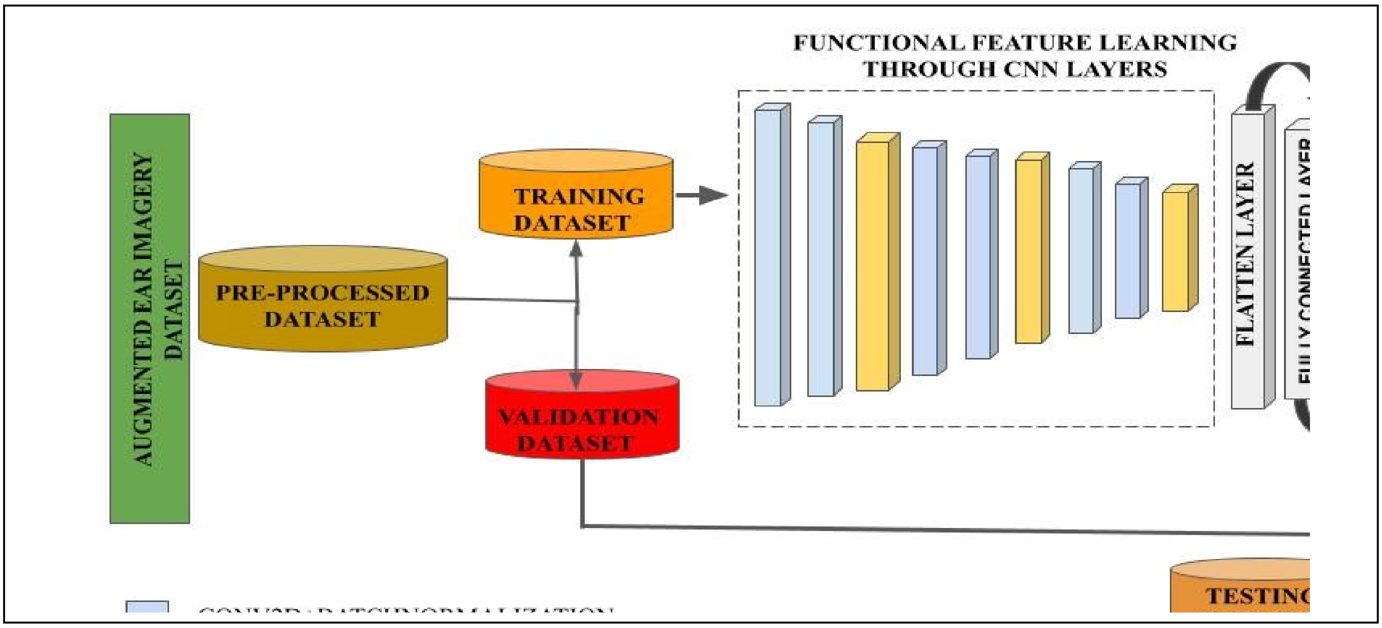

Fig 1. Proposed Method for Ear Condition Detection

They consist of feature maps and neurons in the feature maps extract local characteristics at different positions from the previous layer. The features are obtained by convolving the feature maps with a learned kernel and passing the output to an activation function.
Table I represents the architecture of the proposed model along with the number of parameters.

TABLE I.
CNN ARCHITECTURE EMPLOYED FOR PROPOSED WORK

Layer	**Shape of the Output**	**No. of Parameters**
Conv2D_layer	(None, 273, 247, 32)	896
BatchNormalization_layer	(None, 273, 247, 32)	128
Conv2D_layer_1	(None, 273, 247, 32)	9248
BatchNormalization_layer_1	(None, 273, 247, 32)	128
MaxPool2D_layer	(None, 136, 123, 32)	0
Dropout	(None, 136, 123, 32)	0
Conv2D_layer_2	(None, 136, 123, 64)	18496
BatchNormalization_layer_2	(None, 136, 123, 64)	256
Conv2D_layer_3	(None, 136, 123, 64)	36928
BatchNormalization_layer_3	(None, 136, 123, 64)	256
MaxPool2D_layer_1	(None, 68, 61, 64)	0
Dropout_1	(None, 68, 61, 64)	0
Conv2D_layer_4	(None, 68, 61, 128)	73856
BatchNormalization_layer_4	(None, 68, 61, 128)	512
Conv2D_layer_5	(None, 68, 61, 128)	147584
BatchNormalization_layer_5	(None, 68, 61, 128)	512
MaxPool2D_layer_2	(None, 34, 30, 128)	0
Dropout_2	(None, 34, 30, 128)	0
Flatten	(None, 130560)	0
Dense_layer	(None, 128)	16711808
BatchNormalization_layer_6	(None, 128)	512
Dorpout_3	(None, 128)	0
Dense_layer_1	(None, 4)	516

The CNN architecture used in this work consists of 6 convolutional blocks, and 2 dense layers. The filter size for every convolutional layer is 3×3.
The first convolutional block comprises of a convolutional layer having 32 filters and batch-normalization layer. The second convolutional block comprises a convolutional layer having 32 filters, a batch normalization layer, a max-pooling layer, and a dropout of size 0.2.
The third convolutional block comprises a convolutional layer having 64 filters and a batch normalization layer. The fourth convolutional block comprises a convolutional layer having 64 filters, a batch normalization layer, a max-pooling layer, and a dropout of size 0.3. The fifth convolutional block consists of a convolutional having 128 filters and a batch normalization layer. The sixth convolutional block consists of a convolutional layer having 128 filters, a batch normalization layer, a max-pooling layer, and a dropout of size 0.4. The pool size for max-pooling layers is 2×2. There is a flatten layer after the sixth convolutional block.
The first dense layer consists of 128 neurons and uses ReLu as the activation function. After the first dense layer, there is a batch-normalization layer and dropout having a size of 0.5. The second dense layer consists of 4 neurons corresponding to the four classes and uses Softmax as the activation function. The max-pooling have been incorporated for dimensionality reduction and the drop-outs to prevent the model from overfitting.

III. RESULTS

The experiments were implemented using python employing an Intel Core i5 processor having 8GB RAM and Windows 10 operating system. 24%. of the dataset was used as the testing dataset and 36% of the remaining dataset was for the purpose of validation.

The SGD optimizer to train the model. Its learning rate and momentum of SGD were initialized using values 0.001 and 0.9 respectively. The categorical cross-entropy loss was employed as the loss function. It was trained up to 400 epochs.

The Fig. 3 represents the accuracy and loss plots for the model. Checkpointing was employed to save the model at highest accuracy and it was obtained at 367th epoch. At that point the values of training and validation accuracy were 99.98% and 97.37% respectively.

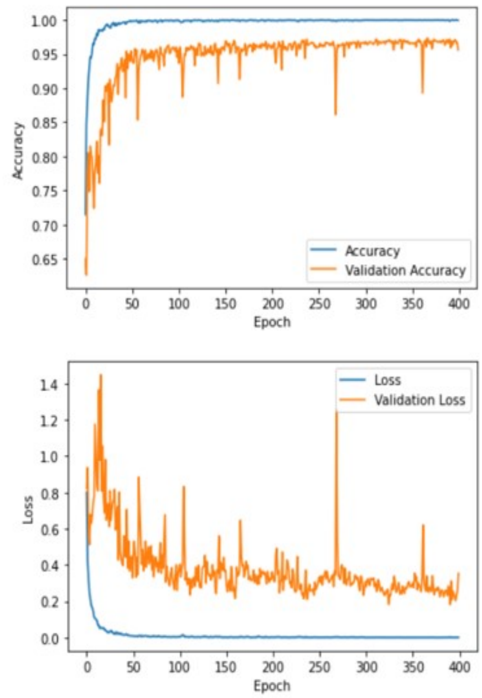

Fig 3. Accuracy and Loss Plots

The trained model was tested using 1369 images. The confusion matrix for the results is shown in Fig 4. The precision, recall and F1-score were used to analyze the performance of the model. Their values for the different classes along with the number of images used for testing are presented in Table II.

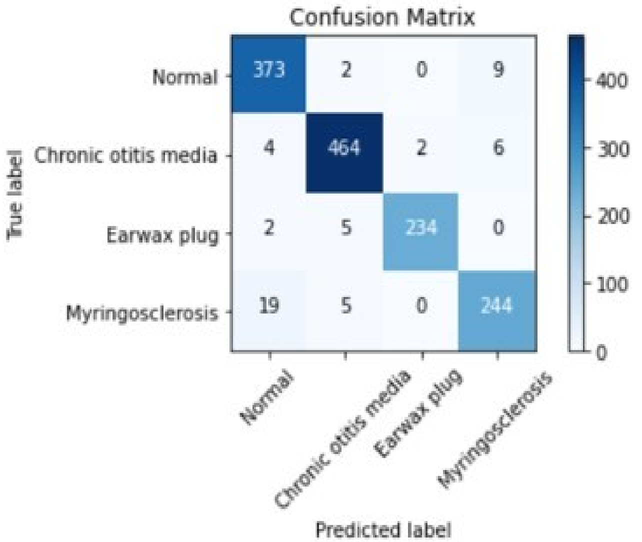

Fig 4. Confusion Matrix

The proposed method obtained a classification accuracy of 96% and outperformed [21], which employed discrete cosine transform and support vector machine for detecting ear conditions using the same dataset and obtained a classification accuracy of 93.9%. It also outperformed [1], which used various CNN for diagnosis of ear conditions and achieved an accuracy of 95%.

TABLE II.
PRECISION, RECALL AND F1 SCORE

Class	Precision	Recall	F1-Score	No. of Images
Normal	0.94	0.97	0.95	384
Chronic Otitis Media	0.97	0.97	0.97	476
Earwax Plug	0.99	0.97	0.98	241
Myringosclerosis	0.94	0.91	0.93	268

IV. CONCLUSION AND FUTURE WORK

A method employing deep learning for detecting earwax plug, chronic otitis media, and myringosclerosis has been proposed in the paper related to disease identification of the ear. It was trained using the ear imagery dataset training and validations accuracies of 99.98% and 97.37% were obtained as a result. It was tested using 1369 images and a classification accuracy of 96% was obtained. The integration of this proposed model can serve as an automatic screening framework for detection of diseases of the ear.

In future work, an effective and reliable device for detecting treatment of ear infections even in the absence of proper medical facilities and staff.

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